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1. **Title of the invention:**
   Smart Wearable Band for Early Prediction of Heat Stroke and Dehydration Using Multimodal Adaptive Digital Twin Prediction (MADTP).

2. **Field /Area of invention:**

   Internet of Things (IoT), Edge Computing, Biomedical Wearable Systems, Industrial Worker Safety, and Machine Learning-driven Predictive Health Diagnostics.

3. **Prior Patents and Publications from literature (provide a table summarizing the prior art)**

   - **US Patent US11375896B2:** Edge-intelligent IoT-based wearable systems utilizing localized inference for health anomaly tracking.
     ***Limitation/Gap*:** Focuses broadly on behavioral or localized neural monitoring without incorporating multi-sensor thermal-fluidic dynamics (such as combined core-skin temperature gradients and galvanic dehydration metrics) or digital twin prognostic modeling.

   - **US Patent US10892543B2:** Wearable hydration and electrolyte monitoring patches utilizing sweat impedance spectroscopy.
     ***Limitation/Gap*:** Restricted exclusively to chemical sweat analysis via direct skin contact patches without integrating ambient microclimate loads (humidity, radiant heat) or predictive machine learning drift extrapolation.

   - **CN Patent CN114512389A:** IoT-enabled smart safety helmet and wearable band integrated with PID-controlled active cooling mechanisms for heat stroke mitigation.
     ***Limitation/Gap*:** Employs reactive on-off or PID hardware cooling actuation based purely on single-point skin thresholds rather than edge-computed digital twin predictive modeling for pre-symptomatic warning.

   - **US Patent US20220295781A1:** Fog and edge computing architecture for real-time physiological data streaming and emergency alert triage.
     ***Limitation/Gap*:** Designed as a generic hospital-to-cloud telemetry framework, lacking lightweight embedded machine learning pipelines (such as Random Forest models optimized for resource-constrained microcontrollers like the ESP32) for autonomous offline edge decision-making.

   - **EP Patent EP3892341A1:** Multi-sensor biometric garment for continuous worker thermal strain evaluation in high-temperature industrial environments.
     ***Limitation/Gap*:** Relies heavily on bulky textile-integrated wiring and centralized server processing, creating significant latency bottlenecks and failing to provide localized multi-modal sensor fusion at the wearable edge.

   - **US Patent US11129584B2:** Dynamic personal thermal comfort prediction system using machine learning classifiers and wearable environmental sensors.
     ***Limitation/Gap*:** Targeted solely at indoor HVAC optimization and general thermal comfort rather than acute medical pathology prediction, emergency heat stroke prevention, or fluid-electrolyte balance degradation tracking.

   - **WIPO Patent WO2023081234A1:** Real-time digital twin synchronization method for human physiological status tracking using wearable IoT telemetry.
     ***Limitation/Gap*:** Implements high-overhead cloud-based synchronization models that require continuous high-bandwidth internet connectivity, making them unreliable for remote industrial outdoor settings or emergency scenarios with intermittent network coverage.

   - **US Patent US20240081657A1:** Smart wearable IoT device integrating multi-modal sensors and hierarchical machine learning modules to track health deterioration.
     ***Limitation/Gap*:** Relies on a split architecture separating shallow on-board processing from heavy cloud-based deep learning, introducing connectivity dependencies and failing to execute a self-contained local digital twin extrapolation loop on edge microcontrollers.

   - **WIPO Patent WO2024050133A1:** Digital twin generation framework for diagnostic and therapeutic use utilizing multi-dataset fusion and real-time data scraping.
     ***Limitation/Gap*:** Geared toward macro-level clinical/genomic diagnostics rather than localized occupational or field-worker wearable monitoring of acute core-skin thermal gradients and hydration shifts.

   - **US Patent US11317805B2:** Wearable health monitoring device utilizing localized sensors for continuous vital sign tracking and trend visualization.
     ***Limitation/Gap*:** Provides static graphical trend readouts without incorporating predictive machine learning models or adaptive digital twin simulation to pre-emptively forecast thermal strain and heat stroke risks.

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4. Summary and background of the invention (Address the gap / Novelty)

Industrial operators working within demanding, high-heat sectors—such as metallurgical foundries, large construction zones, and energy extraction plants—suffer from severe occupational hazards like heat stroke, cardiovascular stress, and sudden physical collapse. Traditional safety gear relies heavily on blunt, fixed thresholds, sounding an emergency alarm only after a worker's heart rate spikes past 140 BPM or body temperature exceeds 38°C. These crude mechanisms fail because they ignore individual biological differences, environmental compounding factors like humidity and ambient temperature, and subtle physical exhaustion trends that build up gradually over long work shifts.

The Novelty: MADTP solves this by building a closed-loop cyber-physical framework that integrates custom edge-wearable hardware with a dedicated local processing server. Instead of static triggers, the platform builds a personalized physiological baseline for every unique operator during an initial calibration phase. It then streams multi-sensor telemetry to run live digital twin divergence analysis and machine learning classification via an optimized Random Forest engine, flagging early signs of physical degradation long before an emergency occurs.

5. Objective(s) of Invention

- To establish an individualized, dynamic physiological baseline for every single worker during a dedicated startup calibration phase, completely neutralizing false alarms caused by natural human health variations.
- To construct a multi-sensor data fusion pipeline that ties together live cardiorespiratory metrics (Heart Rate, SpO2), environmental stressors (heat load index), and motion-based activity heuristics.
- To execute real-time digital twin divergence tracking by continuously comparing live operational telemetry against the worker's unique personal baseline profile.
- To deliver an adaptive risk scoring and automated feedback mechanism that triggers localized buzzer alerts and vibration feedback loops instantly when physical stress crosses safety bounds.

6. Working principle of the invent (in brief)

The system functions through a continuous edge-to-server monitoring loop. The wearable hardware uses non-blocking asynchronous routines (millis()) to poll onboard sensors, verifying proper skin contact via an infrared threshold before formatting clean JSON payloads. These packets are transmitted over Wi-Fi to a local Python Flask server running on port 5050. The server handles strict data validation, freezes a personalized baseline during the initial operating window, extracts rolling feature vectors, and computes both Random Forest deterioration probabilities and digital twin divergence scores to dictate risk categories.

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7. Description of the invention in detail (Include drawing and or photograph as needed)

- Hardware System Composition: Built around an ESP32 microcontroller communicating over a 400kHz fast I2C bus (SDA: 21, SCL: 22). It features a MAX30102 photoplethysmography sensor utilizing circular sample buffers for pulse and SpO2 tracking, a waterproof DS18B20 digital thermal probe for skin surface temperature, a DHT22 module for ambient room conditions, and an MPU6050 6-axis IMU for motion tracking.

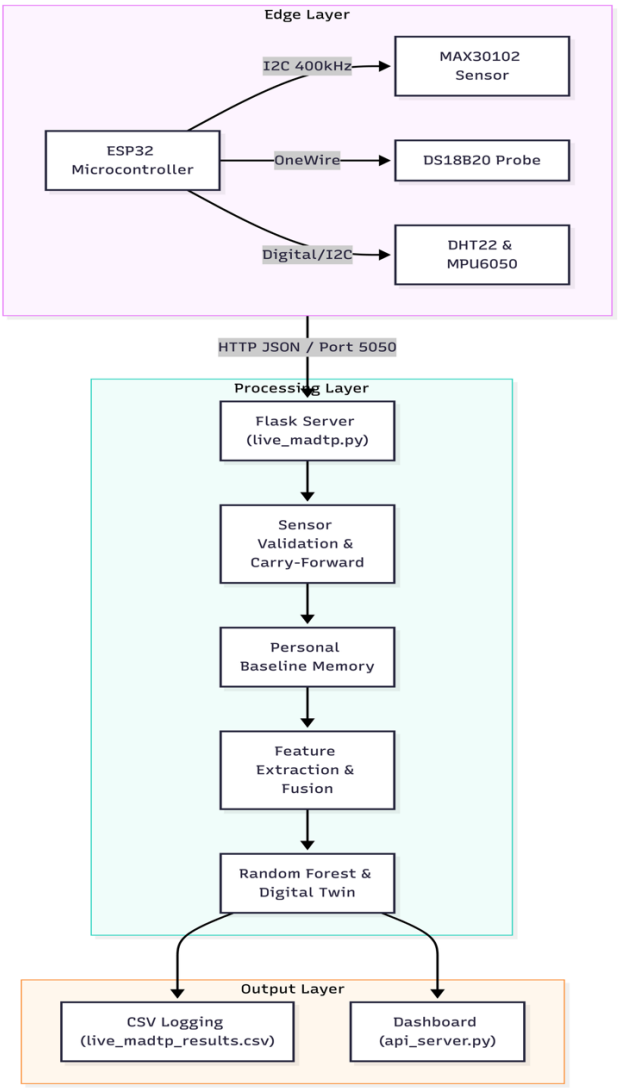


Figure 1 (System Architecture & Data Flow Diagram): This visualizes the physical edge node (ESP32, MAX30102, DS18B20, DHT22, MPU6050) sending structured JSON payloads via Wi-Fi to the Flask ingestion server (live_madtp.py), routing through feature extraction and the Random Forest engine down to local CSV records (live_madtp_results.csv) and the web dashboard (api_server.py).

- Core Mathematical Formulations Developed for the Project:

- Combined Activity Heuristic (A_{combined}): Quantifies physical worker motion by fusing standard deviations across 3-axis accelerometer and gyroscope vector magnitudes:

$$A_{\text{combined}} = 0.7 \cdot \left(\frac{\sigma_{\text{accel_magnitude}}}{2.0}\right) + 0.3 \cdot \left(\frac{\sigma_{\text{gyro_magnitude}}}{250.0}\right)$$

- Environmental Heat Load Index: Evaluates compounding thermal stress by normalizing ambient temperature (T_{ambient}) and relative humidity (Humidity) against safe human tolerance thresholds:

$$\text{Heat Load} = \text{clip}\left(0.55 \cdot \left(\frac{T_{\text{ambient}} - 28.0}{16.0}\right) + 0.45 \cdot \left(\frac{\text{Humidity} - 40.0}{45.0}\right), 0.0, 1.0\right)$$

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- Twin Divergence Index (I_{div}): Measures real-time statistical residual drift between incoming live sensor readings and the worker's frozen personal baseline mean (μ) and standard deviation (σ):

$$I_{div} = \left|\frac{HR_{live} - \mu_{hr}}{\sigma_{hr}}\right| + \left|\frac{SpO2_{live} - \mu_{spo2}}{\sigma_{spo2}}\right| + \left|\frac{Temp_{live} - \mu_{temp}}{\sigma_{temp}}\right|$$

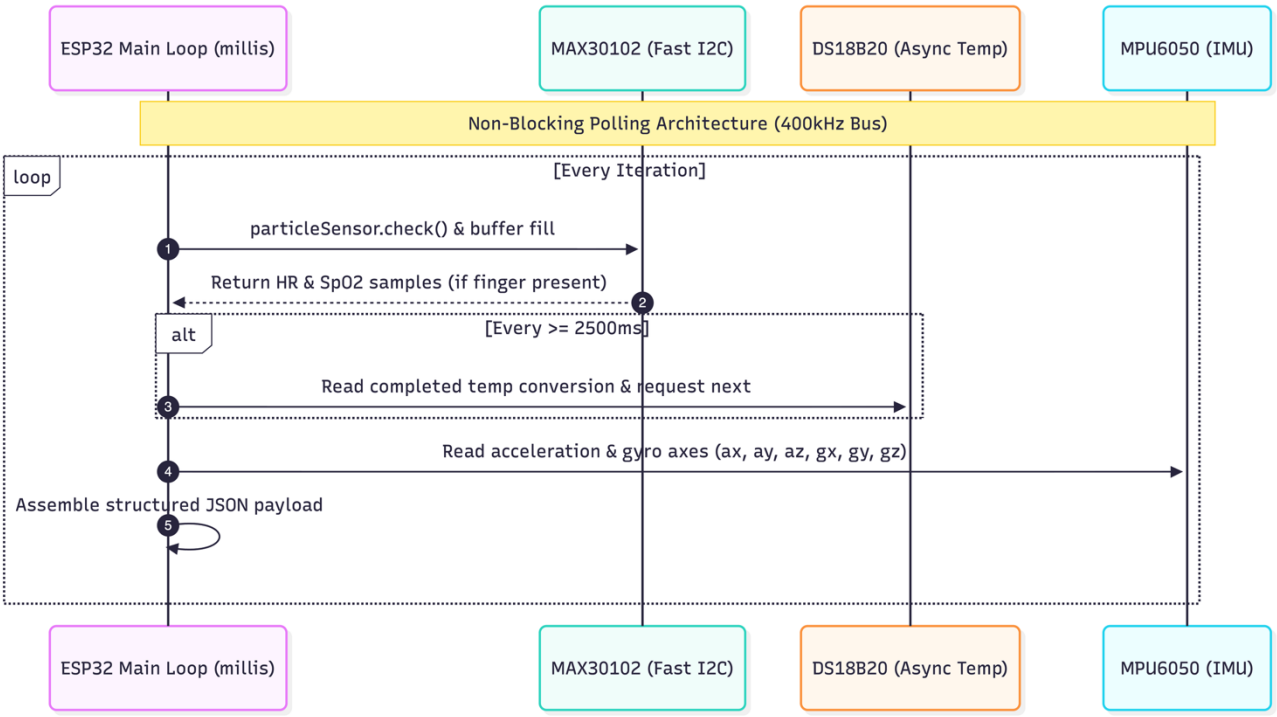
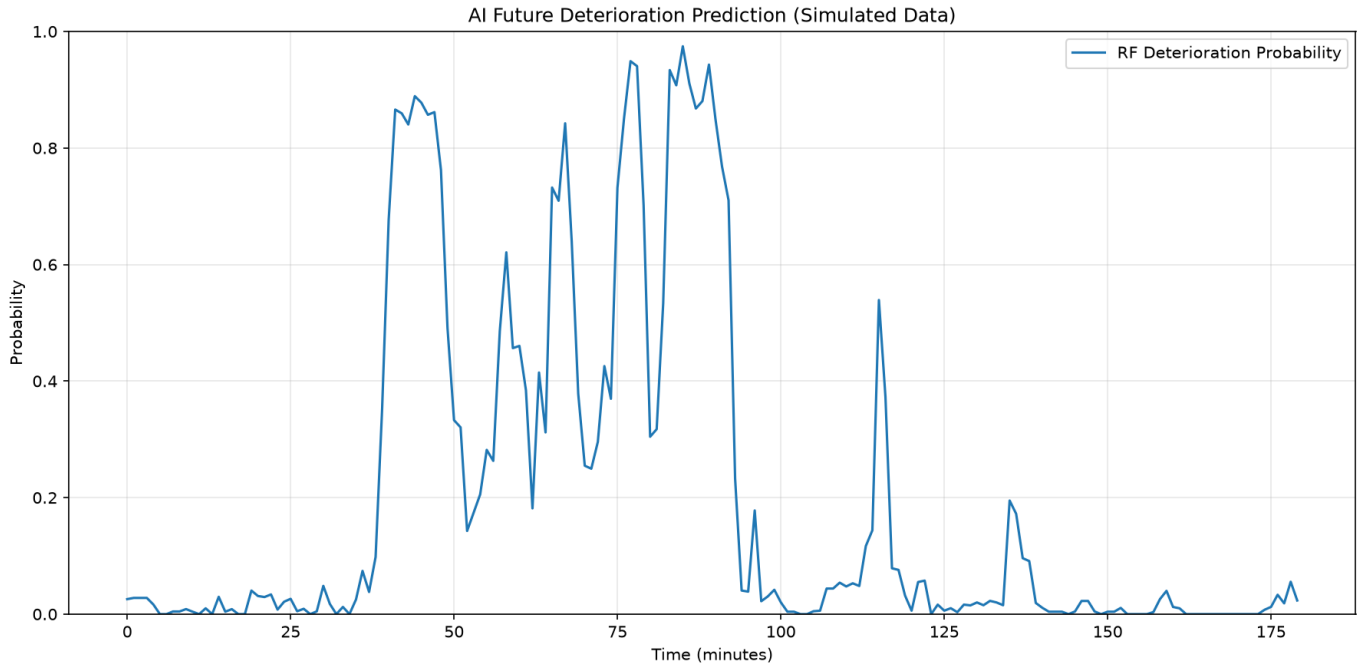


Figure 2 (Hardware Schematic & Timing Sequence): This sequence chart maps out the non-blocking execution loop (millis()), high-speed 400kHz I2C calls, circular buffer checks, asynchronous DS18B20 temperature reads every 2500ms, and final payload assembly.

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8. Experimental validation results

- Empirical Performance Metrics: The backend pipeline demonstrated a consistent operational inference latency of under 50 ms per HTTP packet upload. The system successfully tracked risk progression from safe nominal baselines (~ 0.08 to 0.12) up to acute stress conditions exceeding adaptive safety limits (> 0.50). Furthermore, the software logic successfully managed intermittent packet losses and complete physical finger dropouts by utilizing carry-forward validation windows without crashing server execution.



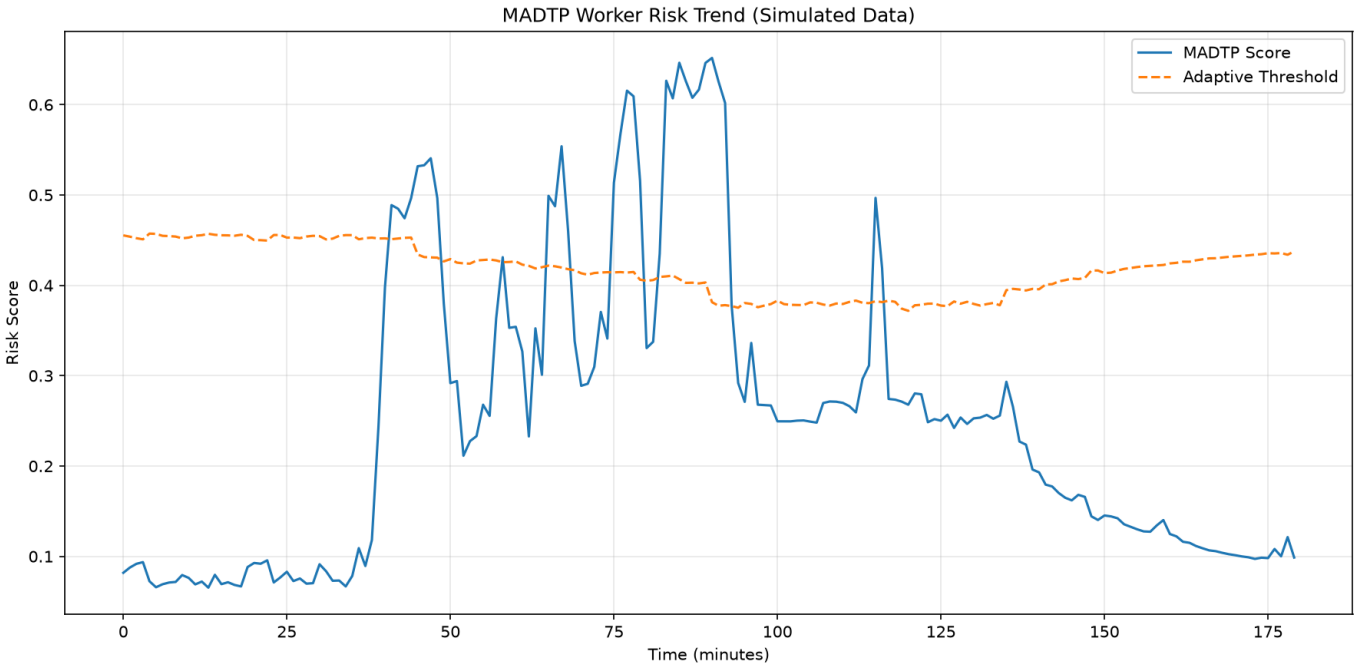


Figure 3 & Figure 4 (Simulation & Risk Analysis Plots): Exhibit personal baseline divergence trends, cumulative heat load compounding, and Random Forest deterioration probability graphs across prolonged operating sessions.

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9. What aspect(s) of the invention need(s) protection?

- The custom sensor validation and carry-forward telemetry filter designed specifically to handle transient finger removal and signal dropouts on resource-limited microcontrollers.
- The unique multimodal fusion mathematical model that combines personal baseline statistical divergence, ambient heat load metrics, and motion activity indices.
- The closed-loop adaptive threshold mechanism that dynamically scales a worker's allowable risk boundaries based on cumulative shift exposure time and surrounding environmental conditions.

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			**Development**			**Deployment**		
TRL 1	**TRL 2**	**TRL 3**	**TRL 4**	**TRL 5**	**TRL 6**	**TRL 7**	**TRL 8**	**TRL 9**
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
✓	✓	✓						

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